

* Machine learning:-

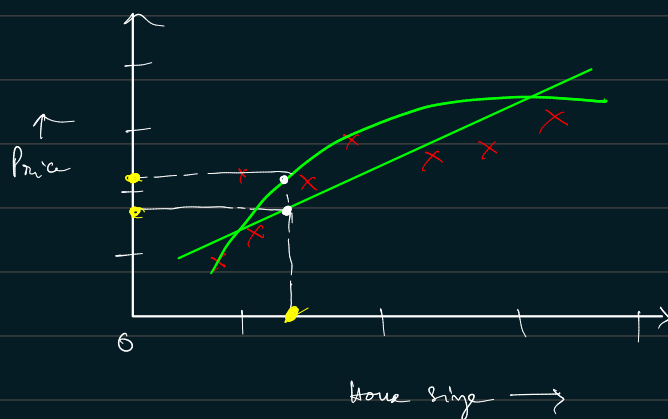
Field of study that gives computers the ability to learn without being explicitly programmed.

* Supervised learning:-

Input	Output	Application
Email	Spam ? (0/1)	Spam filtering
Audio	Text transcripts	Speech recognition
English	Spanish	Machine translation
ad, user info	click ? (0/1)	online advertising
img, radar info	pos. of other cars	self driving car
img of phone	defect ? (0/1)	visual inspection

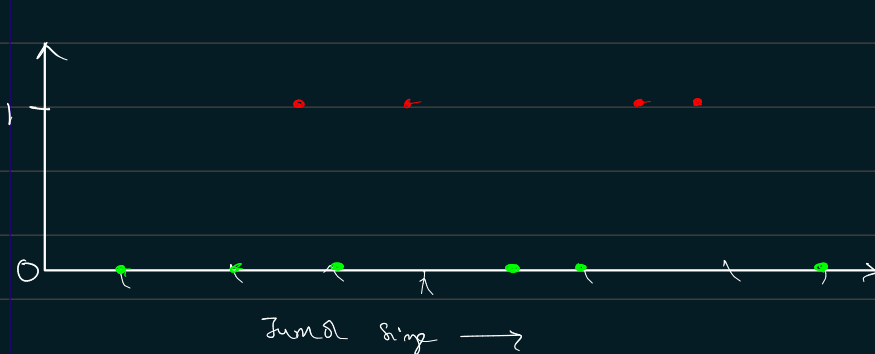
* Regression:-

Predict a number from infinitely many possible outputs.



* Classification:-

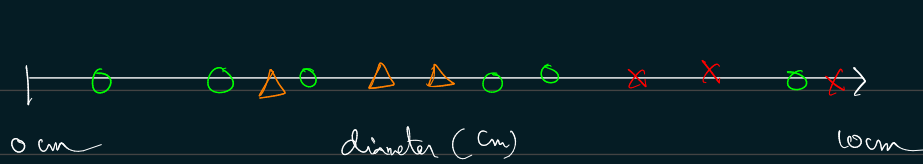
Ex Breast Cancer detection



Size	diagnosis
2	0
5	1
7	0
3	1
1	0

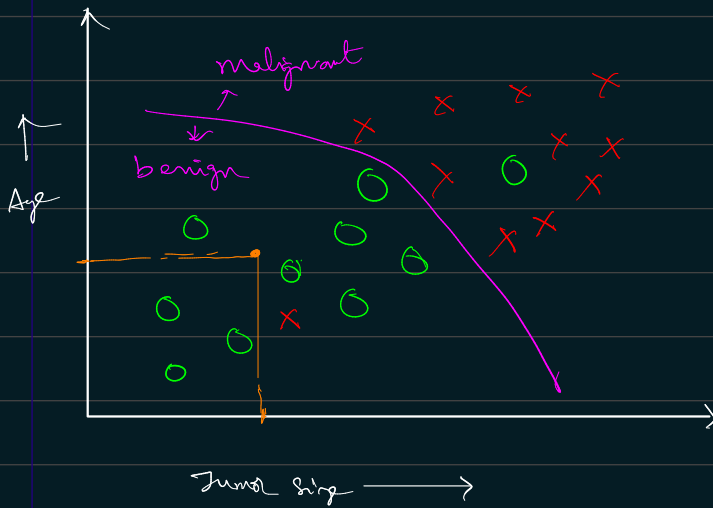
malignant benign





○ benign
 × malignant type 1
 △ malignant type 2

Two more inputs :-

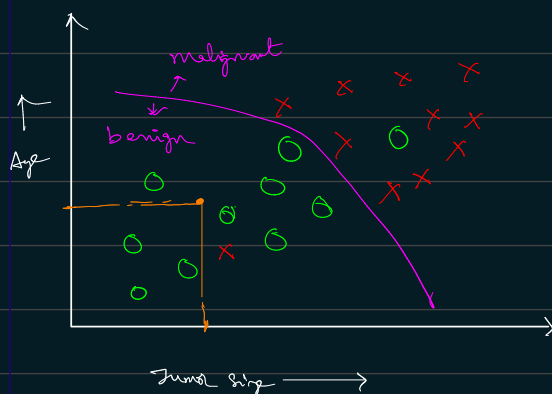


* Support vector machine :-

An algorithm which allows to use ∞ inputs (∞ dimensional vectors).

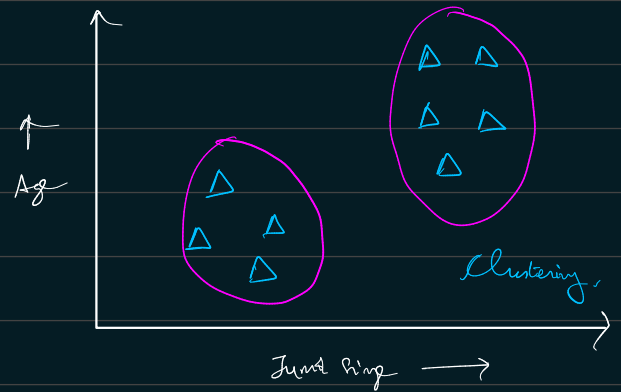
Kernels.

* Unsupervised learning :-



Supervised learning

→ learns from data labeled with "right answers"



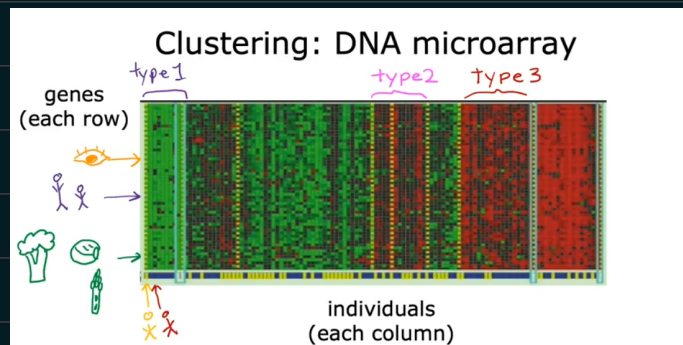
Unsupervised learning

→ Finds something interesting in unlabeled data

Ex:-

Clustering is used in Google news to recommend related news

Ex:-



* Unsupervised learning :-

Data only comes with input x , but not output labels y .

Algorithm has to find structure in the data

Clustering :-

Group similar data points together

Anomaly detection

Find unusual datapoints

Dimensionality reduction

Compress data using fewer numbers.